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Evaluating the Safety and Efficacy of Asundexian in Cardiovascular Disease: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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Abstract

Background: Thromboembolic events remain a major cause of morbidity and mortality in cardiovascular diseases, particularly myocardial infarction, atrial fibrillation (AF), and ischemic stroke. While standard anticoagulants reduce these events, their use is limited by bleeding risks. Asundexian, a Factor XIa inhibitor, has emerged as a potential agent to prevent thrombosis while preserving hemostasis. This meta-analysis evaluates its safety and efficacy across cardiovascular conditions.

Methods: Following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, a systematic review and meta-analysis was registered with PROSPERO (CRD420251039715). Randomized controlled trials (RCTs) comparing Asundexian with placebo or active controls were identified via PubMed, Scopus, and CENTRAL. Outcomes included all-cause mortality, cardiovascular death, ischemic stroke, stroke or systemic embolism, major/clinically relevant non-major bleeding, and adverse effects. Data were analyzed using a random-effects model; study quality was assessed via Cochrane Risk of Bias Tool 2.0.

Results: Four RCTs were included. The pooled analysis showed no significant difference in all-cause mortality between Asundexian and control (RR 1.04, 95% CI 0.73-1.48, $P = .83$), with low heterogeneity ($I^2 = 13\%$). Ischemic stroke rates were also similar (RR 1.64, 95% CI 0.51-5.25, $P = .40$), but with high heterogeneity ($I^2 = 89\%$). Cardiovascular-related mortality did not differ between groups (RR 1.13, 95% CI 0.77-1.66, $P = .53$; $I^2 = 0\%$). For safety outcomes, ISTH major or clinically relevant non-major bleeding showed a non-significant trend toward lower risk with Asundexian (RR 0.63, 95% CI 0.21-1.91, $P = .41$; $I^2 = 77\%$). Stroke or systemic embolism (RR 1.29, 95% CI 0.08-20.70, $P = .88$) and adverse events (RR 1.00, 95% CI 0.95-1.04, $P = .85$) were comparable between groups.

Conclusion: Asundexian demonstrated no significant difference compared to control in all-cause mortality, ischemic stroke, or cardiovascular-related mortality. While there was a non-significant trend toward reduced bleeding risk, no clear benefit was observed for stroke prevention or systemic embolism. The safety profile was comparable, with no increase in adverse events. These findings suggest that Asundexian, at the studied doses, does not provide significant clinical advantages in mortality or thromboembolic outcomes but may have a favorable bleeding profile. Further research is needed to define its potential role in specific high-risk populations.

Keywords

atrial fibrillation, factor XIa inhibitors, asundexian, cardiovascular disease

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Introduction

Thromboembolic illness remains a primary cause of morbidity and death throughout the world, particularly with respect to cardiovascular disease such as atrial fibrillation

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(AFib), ischemic stroke, and acute myocardial infarction (AMI).[1] These diseases are highly linked with deranged clotting of the blood in the circulatory system and have the potential to cause life-threatening complications.[2] Anticoagulant therapy has long been at the heart of the prevention and control of disease, as well as thrombotic event prevention.[3] But traditional anticoagulants such as vitamin K antagonists and direct oral anticoagulants (DOACs) carry with them a substantial risk of bleeding, which has raised clinical decision-making issues, particularly in patients with high or elderly risk profiles.[4] This has prompted the pursuit of more selective anticoagulant approaches designed to dissociate thrombotic risk reduction from uncontrolled bleeding liability.

An essential enzyme in the coagulation cascade's intrinsic pathway is factor XI (FXI). When FXIa is activated, it increases the production of thrombin by activating Factor IX, which results in the creation of fibrin clots. Unlike Factor XII (FXII), which initiates the contact activation pathway and is not essential for normal hemostasis, FXI plays a more central role in pathologic thrombosis.[5] Although spontaneous bleeding is uncommon in people with FXI deficiency, there is genetic and clinical evidence suggesting they have decreased incidence of venous thromboembolism, stroke, and myocardial infarction. In contrast, major bleeding disorders are frequently caused by deficits in other coagulation factors. As a result, FXI has become a particularly appealing therapeutic target, providing the potential to interfere with thrombosis without significantly impairing physiological hemostasis, unlike medicines that operate on downstream targets such as thrombin or Factor Xa.[6] Compared to FXII or conventional targets in the coagulation cascade, FXI inhibition is an alternative approach since it can provide antithrombotic efficacy with a lower risk of bleeding.

Asundexian is a first-in-class, oral small molecule that selectively inhibits activated Factor XI (Factor XIa), a new target in the intrinsic coagulation pathway.[7,8] In comparison with agents acting on multiple components in the clotting cascade, the specificity of Asundexian can best preserve normal hemostasis and block pathologic thrombosis in the optimal way. As a Factor XIa inhibitor, Asundexian disrupts thrombus formation during a more advanced phase of the coagulation process but preserves homeostasis mechanisms required to prevent pathologic bleeding.[9,10] The rationale for the targeting of Factor XIa is both mechanistic and clinical, notably in patients with congenital Factor XI deficiency, who have a reduced frequency of thromboembolic complications but, in turn, infrequently present with spontaneous bleeding. This profile indicates that Factor XIa has a limited role in physiological clotting but is disproportionately engaged in pathologic thrombosis.[11,12] Under these circumstances, Asundexian has attracted interest as a potential candidate for use in a wide range of cardiovascular diseases. In AFib, loss of synchronized atrial contraction promotes blood stasis and thrombus formation and significantly raises the risk of stroke.[13] Likewise, ischemic stroke most commonly occurs because of cerebral artery occlusion by thrombi derived from cardiac or arterial sources.[14,15] AMI is usually caused by thrombotic coronary artery obstruction.[16] In each of these circumstances, the capability to separate antithrombotic effectiveness from bleeding hazard constitutes a major progress in cardiovascular medicine. Asundexian, with its targeted approach to inhibiting Factor XIa, can potentially meet this unmet clinical need, providing a new therapeutic option specifically suited to the complex needs of high-risk cardiovascular patients. Its selective inhibition of Factor XIa makes it a strong candidate in the developing field of anticoagulant treatment of cardiovascular disease.

Among FXIa inhibitors, Asundexian was selected for this review due to its promising pharmacological profile, oral administration, and availability of early clinical data across multiple indications. Although other agents like Milvexian and Abelecimab are also being studied, Asundexian has shown potential in preliminary trials, warranting a focused review to better understand its emerging role in anticoagulation. Recent clinical investigations have started to substantiate the conceptual strengths of Factor XIa inhibition in practice. Early-phase trials have shown that Asundexian accomplishes substantial declines in coagulation activity without significantly increasing the risk of major or clinically relevant non-major bleeding, a balance long sought after in anticoagulant therapy.[17] This harm-benefit analysis is especially strong in susceptible patients, like those with recent ischemic stroke, non-cardioembolic stroke, or receiving percutaneous coronary intervention (PCI), where the risks of bleeding can outweigh the benefits of conventional anticoagulation.[18] Additionally, Asundexian's once-daily oral administration and beneficial pharmacokinetic characteristics, such as a long half-life and low drug-drug interactions, reinforce its clinical accessibility and adherence potential.[19] As evidence advances with late-phase randomized controlled trials, the emerging evidence could cement Factor XIa blockade as a new paradigm in antithrombotic therapy that transforms the safety-efficacy balance. Therefore, Asundexian not only represents a mechanistic advance but may also herald an evolution in thromboembolic prevention standards for cardiovascular disease.

The purpose of this systematic review and Meta-Analysis (SRMA) is to evaluate the safety and effectiveness of Asundexian in patients with different cardiovascular disorders. This study aims to provide a strong evidence base to support Asundexian's clinical integration and help improve treatment approaches by systematically reviewing existing literature.

Methodology

This meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses

(PRISMA) criteria and registered with PROSPERO 2025 CRD420251039715. Accessible at https://www.crd.york.ac.uk/PROSPERO/view/CRD420251039715

Search Strategy and Data Sources

This meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.[20] An extensive literature search on PubMed (MEDLINE), Scopus, and the Cochrane CENTRAL Register of Controlled Trials (CENTRAL) was conducted, from inception till May 2025. Search Strategies incorporated a combination of MeSH terms and keywords such as "Cardiovascular Diseases" OR "Myocardial Infarction" OR "Stroke" OR "Atrial Fibrillation" OR "Thromboembolism" OR "Ischemic Stroke" OR "Myocardial Ischemia" OR "Myocardial Infarction" OR "Vascular Diseases" AND "asundexian" OR "Benzamides" OR "Factor XIa inhibitor" OR "coagulation factor XIa". Additionally, we performed manual searches of reference lists from all eligible studies, review articles, and prior meta-analyses to identify potentially relevant studies.

Selection of Studies

Studies were eligible for inclusion if they were randomized controlled trials (RCTs) that evaluated the efficacy or safety of Asundexian in adult patients with cardiovascular diseases, including myocardial infarction (MI), stroke, or atrial fibrillation (Afib). Eligible comparators included apixaban, placebo, or other anticoagulants. Only studies that matched all components of this eligibility framework were included in the analysis. Observational studies, including case-control, cohort (retrospective and prospective), and cross-sectional studies, were excluded. Additionally, articles such as commentaries, editorials, and case reports were also not considered for inclusion.

Data Extraction and Risk of Bias Assessment

All retrieved records were imported into Rayyan, and duplicates were removed. Five reviewers independently screened titles and abstracts, followed by full-text assessment of potentially eligible studies. Any disagreement regarding interpretation and individual study inclusion was resolved by discussion before the final analysis. The study selection process is summarized in the PRISMA flowchart (Figure 1). Data extraction was carried out by two independent reviewers (SM and MA), and any discrepancy was resolved by discussion or a third investigator. Risk of bias was assessed by two independent investigators using the Cochrane Risk of Bias 2.0 (RoB 2) tool for all four eligible RCTs, and the results were matched. Any discrepancy was resolved by discussion and a third reviewer. A Microsoft Excel spreadsheet was created to compile baseline characteristics and outcome data extracted from the included RCTs. Important Baseline characteristics across studies included: number of males and females, follow-up time, mean age (years), race (white, black, Asian, American Indian or Alaska Native, multiple, and others), geographical region (Eastern Europe, North America, South America, Asia, Western Europe, Australia, or Israel), Bodyweight (kg), BMI (kg/m²), CHA2DS2-VASc score, and comorbidities (Hypertension, Hyperlipidemia, Heart failure, Coronary artery disease, Peripheral artery disease, Diabetes, Chronic kidney disease, Carotid endarterectomy or stenting, Myocardial infarction, Anemia, Stroke or transient ischemic attack, PCI or Cabg)

Outcome Assessment

The primary efficacy outcomes were ischemic stroke and all-cause mortality. Secondary efficacy outcomes included death from cardiovascular causes and stroke or systemic embolism. Safety outcomes comprised ISTH-defined major or clinically relevant nonmajor bleeding and any adverse events (AEs). All outcomes were analyzed according to the definitions provided in the original trials.

Statistical Analysis

All statistical analyses were conducted using the latest version of Review Manager (RevMan 5, The Cochrane Collaboration). Results were presented using forest plots to visually synthesize findings. As all outcomes were dichotomous, risk ratios (RRs) with corresponding 95% confidence intervals (CIs) were calculated using a random-effects model. Statistical heterogeneity was assessed using the I^2 statistic. Although the included studies enrolled patients with varying cardiovascular conditions (atrial fibrillation, myocardial infarction, or ischemic stroke), no formal subgroup analyses were performed due to the limited number of studies. Instead, findings were interpreted in the context of disease-specific populations where applicable, based on study-level characteristics.

Publication Bias

Because of the limited number of RCTs (4) included in our analysis, we could not evaluate publication bias for each outcome effectively. The limited sample size compromised the validity of both funnel plot interpretation and statistical testing for bias detection.

Ethics Statement

This meta-analysis was conducted in accordance with the PRISMA guidelines. As it involved the analysis of previously published data and did not include any direct involvement of human participants or access to individual patient data, institutional review board (IRB) or ethics committee approval was not required.

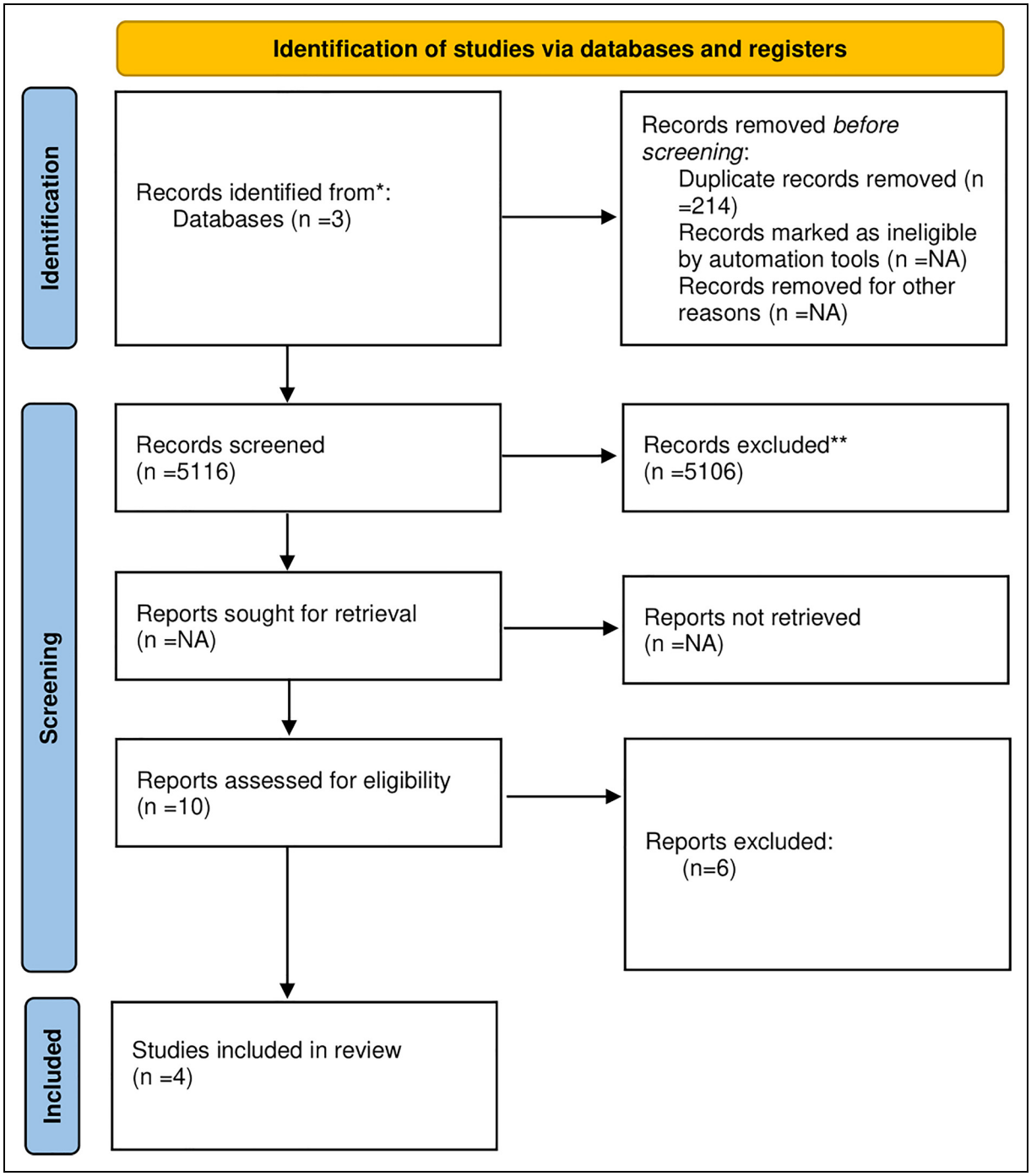

Figure 1. PRISMA Flowchart. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Results

Study Identification

A total of 5213 articles were identified through various database searches. After removing 214 duplicate records, 5002 articles remained for screening. Title and abstract screening led to the exclusion of 4992 articles. The full texts of the remaining 10 articles were assessed for eligibility, of which 6 were excluded for not meeting the inclusion criteria. Ultimately, four randomized controlled trials were included in the final analysis. PRISMA flow chart given in Figure 1.

Characteristics of Studies

All four included articles[21–24] were randomized controlled trials done on the patients with any cardiovascular disorder

that were either given asundexian as intervention or placebo or apixaban as comparator. The meta-analysis was conducted on a combined sample of 17 020 patients, with 8518 receiving 50 mg of Asundexian and 8502 receiving either placebo or apixaban as control. (Table 1) provides a comprehensive summary of the baseline characteristics and study details for the included studies.

Risk of Bias Assessment

To evaluate the risk of bias in the included studies, we employed the Cochrane Risk of Bias Tool (RoB 2).[25] Two reviewers independently assessed each study, and discrepancies were discussed and resolved with a third reviewer. Bias was assessed using the RoB 2 tool across five domains. They rated each domain as "low risk," "some concerns" or "high risk" of bias (Figure 2).

Our risk-of-bias evaluation revealed that while most domains such as outcome measurement, missing data, deviations from intended interventions, and the randomization process were assessed as low risk, the domain related to selection of the reported result was flagged with "some concerns." Consequently, the overall bias assessment for some trials reached "some concerns." According to Cochrane and GRADE guidance, studies with "some concerns" may introduce plausible bias that can subtly skew effect estimates, especially if they contribute substantially to the pooled results In particular, selective outcome reporting may lead to overestimated efficacy or under-reported harms, thereby inflating confidence in Asundexian's effects. Therefore, the current pooled estimates should be interpreted with caution.

Efficacy Measures and Outcome Analysis

Primary Outcomes

All-Cause Mortality. Figure 3 presents a forest plot of four studies assessing the effect of Asundexian compared to control on all-cause mortality. The pooled Risk Ratio was 1.04 (95% CI: 0.73-1.48; $P = .83$), suggesting no statistically significant difference in mortality rates between the groups. At the study level, Piccini (2022) reported an RR of 0.98 (95% CI: 0.26-3.8), Rao (2022) showed an RR of 1.43 (95% CI: 0.55-3.71), Shoamanesh (2022) reported an RR of 1.73 (95% CI: 0.80-3.76), and Piccini (2025) found an RR of 0.84 (95% CI: 0.60-1.19). None of these were statistically significant. Heterogeneity was low ($I^2 = 13\%$, Chi^2 test ($P = .33$), and the overall Z-test value was 0.21 ($P = .83$), confirming the absence of significant difference in all-cause mortality between the two groups.

Ischemic Stroke. Figure 4 illustrates the forest plot comparing ischemic stroke incidence between Asundexian and control groups across four studies. The pooled RR was 1.64 (95% CI: 0.51-5.25; $P = .40$), indicating no statistically significant difference. At the study level, Piccini (2022) reported an RR of 2.95 (95% CI: 0.12-72.14), Rao (2022) reported 0.20 (95% CI: 0.01-4.14), Shoamanesh (2022) found 1.08 (95% CI: 0.81-1.97), and Piccini (2025) showed 4.04 (95% CI: 2.51-6.80). These results reflect a wide range of estimates with overlapping confidence intervals, highlighting inconsistency. Heterogeneity was substantial ($I^2 = 89\%$, Chi^2 test ($P < .00001$), suggesting significant variation across studies. The Z-test statistic for the overall effect was 0.83 ($P = 0.40$), indicating no statistically significant reduction in ischemic stroke risk with Asundexian.

Secondary Outcomes

Cardiovascular Related Mortality. Figure 5 displays a forest plot comparing cardiovascular mortality between Asundexian and control groups across three studies. The pooled Risk Ratio (RR) was 1.13 (95% CI: 0.77-1.66; $P = .53$), indicating no statistically significant difference in cardiovascular mortality. Individually, Piccini (2022) reported an RR of 0.98 (95% CI: 0.20-4.83), Rao (2022) reported 2.49 (95% CI: 0.49-12.78), and Piccini (2025) reported 1.09 (95% CI: 0.72-1.64), with none of the studies showing significant effects. Heterogeneity was negligible ($I^2 = 0\%$, $Chi^2 = 0.96$; $P = .62$), and the overall Z-test for effect yielded $P = .53$, confirming the absence of a statistically significant difference in cardiovascular mortality between groups.

ISTH major or Clinically Relevant non-major Bleeding. Figure 6 presents a forest plot assessing the incidence of ISTH major bleeding events between Asundexian and control groups. The pooled RR was 0.63 (95% CI: 0.21-1.91; $P = .41$), suggesting a non-significant trend favoring Asundexian. Study-wise, Piccini (2022) reported an RR of 0.20 (95% CI: 0.02-1.67), Shoamanesh (2022) reported 1.76 (95% CI: 0.69-4.40), and Piccini (2025) reported 0.44 (95% CI: 0.34-0.67). Although the latter showed a numerically lower risk, the overall result was not statistically significant. Heterogeneity was considerable ($I^2 = 77\%$, $Chi^2 = 8.64$; $P = .01$), and the Z-test for effect showed $P = .41$, indicating no clear difference in major bleeding rates.

Occurrence Stroke or Systemic Embolism. Figure 7 displays the forest plot for stroke or systemic embolism comparing Asundexian and control groups. The pooled RR was 1.29 (95% CI: 0.08-20.70; $P = 0.88$), showing no statistically significant difference and wide confidence intervals reflecting limited event data. Piccini (2022) could not be estimated due to zero events in both arms, Rao (2022) reported an RR of 0.20 (95% CI: 0.01-4.14), and Piccini (2025) reported 3.76 (95% CI: 2.44-5.79). Heterogeneity was substantial ($I^2 = 72\%$, $Chi^2 = 2.85$;

Table 1. Baseline Characteristics of Studies.

Studies	J.P. Piccini, et al 2025	Jonathan P Piccini, et al 2022	Ashkan Shoamanesh, et al 2022	Sunil V. Rao, et al 2022
Study type	RCT	RCT	RCT	RCT
Population	Atrial Fibrillation	Atrial Fibrillation	Stroke	Acute MI
Intervention	Asundexian, 50 mg	Asundexian, 20 and 50 mg	Asundexian, 10, 20 and 50 mg	Asundexian, 10, 20 and 50 mg
Control	Apixaban	Apixaban	Placebo	Placebo
Total Patients:				
Asundexian total:	14 810	755	1808	1601
10 mg	-	-	455	397
20mg	-	251	450	401
50mg	7415	254	447	401
Control	7395	250	456	401
Males:				
Asundexian:				
10 mg	-	-	294	-
20 mg	-	148	300	-
50 mg	-	157	293	-
Control	-	141	306	-
Females:				
Asundexian:				
10 mg	-	-	161	93
20 mg	-	148	150	87
50 mg	2656	157	154	100
Control	2558	141	150	90
Follow up time	155 days	12 weeks	26-52 weeks	368 days
Age (year):				
Asundexian:				
10 mg	-	-	66.8 ± 10.1	67.0 (62,73)
20 mg	-	73.6 ± 8.0	67.6 ± 9.4	68(61, 73)
50 mg	73.9 ± 7.7	73.1 ± 8.5	67.0 ± 10.0	68 (63,73)
Control	73.9 ± 7.7	74.3 ± 8.3	66.6 ± 10.1	68(60,73)
White:				
Asundexian:				
10 mg	-	-	381	334
20 mg	-	211	377	345
50 mg	5216	212	367	347
Control	5211	209	380	339
Black:				
Asundexian:				
10 mg	-	-	4	3
20 mg	-	1	3	2
50 mg	88	1	8	2
Control	95	1	3	1
American Indian or Alaska Native:				
Asundexian:				
10 mg	-	-	-	1
20 mg	-	-	-	2
50 mg	-	-	-	0
Control	-	-	-	0
Multiple races:				
Asundexian:				
10 mg	-	-	-	1
20 mg	-	-	-	0
50 mg	-	-	-	2
Control	-	-	-	2
Body weight (kg):				

(continued)

Table 1. Continued.

Studies	J.P. Piccini, et al 2025	Jonathan P Piccini, et al 2022	Ashkan Shoamanesh, et al 2022	Sunil V. Rao, et al 2022
Asundexian:				
10 mg	-	-	78.5 ± 17.4	80 (70, 91)
20 mg	-	-	78.6 ± 16.4	80(70,91.6)
50 mg	-	-	79.2 ± 16.5	80(72,94)
Control	-	-	79.2 ± 16.0	80.5 (70,92)
BMI, kg/m²				
Asundexian:				
10 mg	-	-	27.4 ± 5.0	-
20 mg	-	-	27.4 ± 4.8	-
50 mg	-	-	27.6 ± 4.9	-
Control	-	-	27.4 ± 4.7	-
Previous DOAC use				
Asundexian:				
10 mg	-	-	-	-
20 mg	-	109	-	-
50 mg	1238	116	-	-
Control	1255	111	-	-
CHA2DS2-VASc score				
Asundexian:				
10 mg	-	-	-	-
20 mg	-	3.9 ± 1.4	-	-
50 mg	4.3 ± 1.3	3.8 ± 1.3	-	-
Control	4.3 ± 1.3	4.1 ± 1.4	-	-
Hypertension				
Asundexian:				
10 mg	-	-	366	275
20 mg	-	226	333	278
50 mg	6558	227	347	282
Control	6565	220	346	292
Hyperlipidemia				
Asundexian:				
10 mg	-	-	225	-
20 mg	-	142	228	-
50 mg	4747	153	256	-
Control	4719	152	246	-
Heart Failure				
Asundexian:				
10 mg	-	-	28	-
20 mg	-	108	26	-
50 mg	3456	107	26	-
Control	3473	117	24	-
Coronary artery disease				
Asundexian:				
10 mg	-	-	43	172
20 mg	-	76	41	174
50 mg	2496	71	46	160
Control	2452	85	37	163
Peripheral artery disease				
Asundexian:				
10 mg	-	-	11	31
20 mg	-	-	9	29
50 mg	-	-	12	22
Control	-	-	13	29
Diabetes				
Asundexian:				
10 mg	-	-	122	166

(continued)

Table 1. Continued.

Studies	J.P. Piccini, et al 2025	Jonathan P Piccini, et al 2022	Ashkan Shoamanesh, et al 2022	Sunil V. Rao, et al 2022
20 mg	-	83	137	154
50 mg	2722	74	115	158
Control	2748	87	127	170
Chronic Kidney Disease				
Asundexian:				
10 mg	-	-	19	28
20 mg	-	55	29	13
50 mg	1399	84	16	25
Control	1357	77	29	32
Stroke or TIA				
Asundexian:				
10 mg	-	-	77	23
20 mg	-	22	68	18
50 mg	1389	18	65	26
Control	1305	25	75	20
Myocardial Infarction				
Asundexian:				
10 mg	-	-	23	-
20 mg	-	26	18	-
50 mg	-	41	23	-
Control	-	36	17	-
Previous PCI or CABG				
Asundexian:				
10 mg	-	-	16	-
20 mg	-	60	23	-
50 mg	-	62	22	-
Control	-	60	19	-

DOAC, direct oral anticoagulant; PCI, percutaneous coronary intervention.

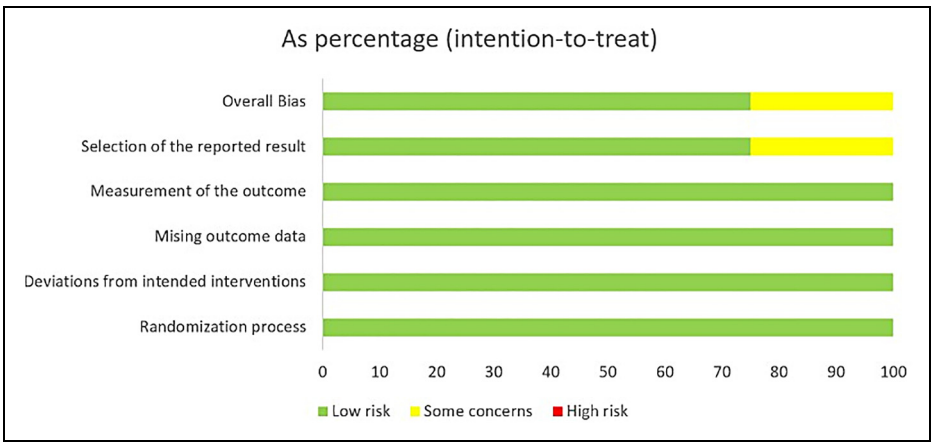

Figure 2. Risk of Bias Assessment.

$P = 0.06$), and the overall Z-test P-value was 0.88, indicating no statistically significant effect on stroke or systemic embolism.

Any Adverse Events. Figure 8 displays a forest plot comparing the incidence of any adverse events between Asundexian and control groups across three studies. The pooled Risk Ratio (RR) was 1.00 (95% CI: 0.95-1.04; $P = .85$), indicating no statistically significant difference in ADEs between groups. Study-wise, Piccini (2022) reported an RR of 0.97 (95% CI: 0.81-1.18), Rao (2022) reported 0.88 (95% CI: 0.69-1.33), both showing no significant difference, while Piccini (2025) reported an RR of 1.00 (95% CI:

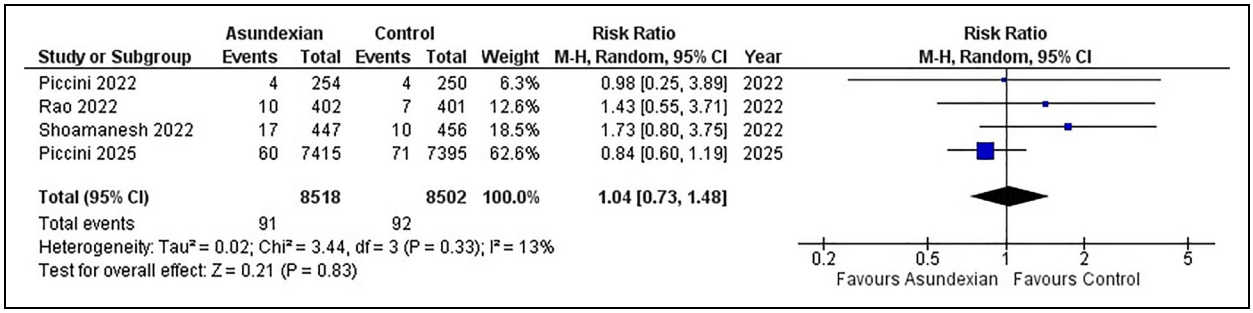

Figure 3. Forest Plot of all-Cause Mortality.

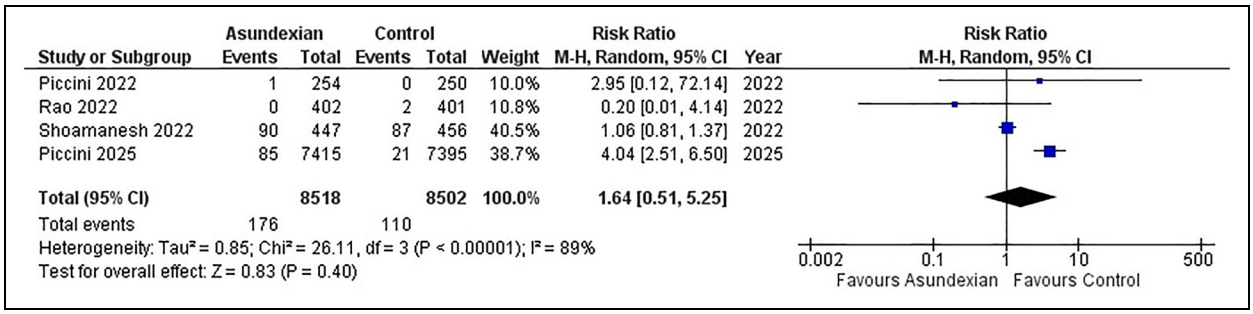

Figure 4. forest Plot of Ischemic Stroke.

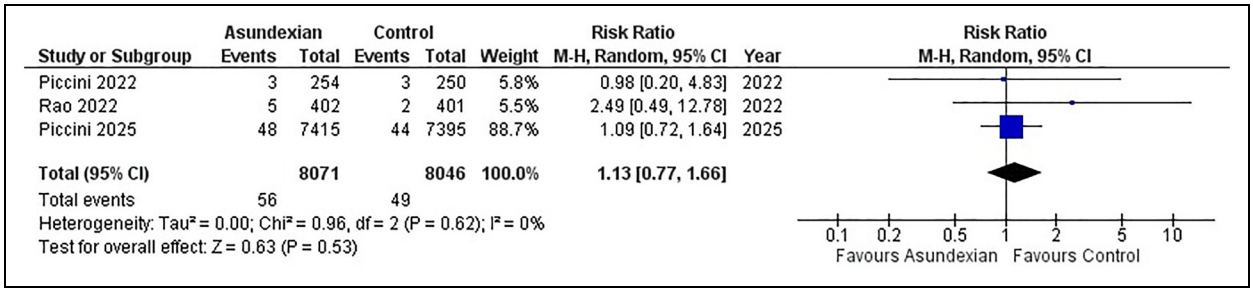

Figure 5. Forest Plot of Cardiovascular Related Mortality.

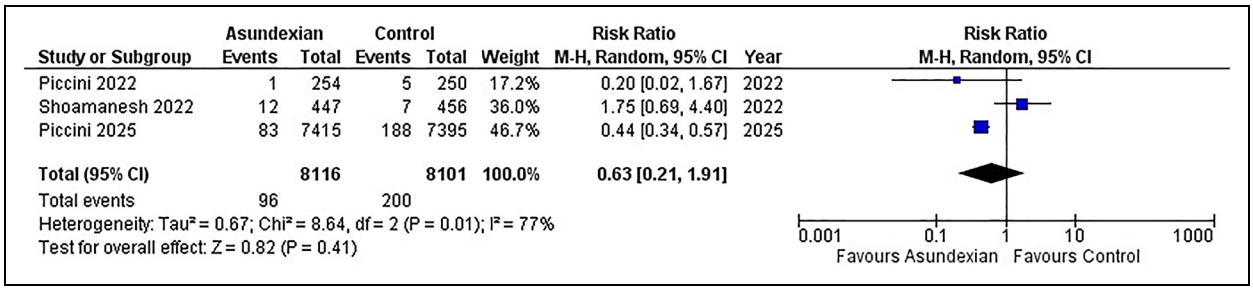

Figure 6. Forest Plot of ISTH Major or Clinically Relevant Non-Major Bleeding.

0.96-1.04), also indicating no significant effect. Heterogeneity among the studies was negligible ($I^2 = 0\%$, Chi² test ($P = .80$), and the overall Z-test for effect was 0.19 ($P = .85$), confirming the lack of a statistically significant difference in ADE rates between Asundexian and control groups.

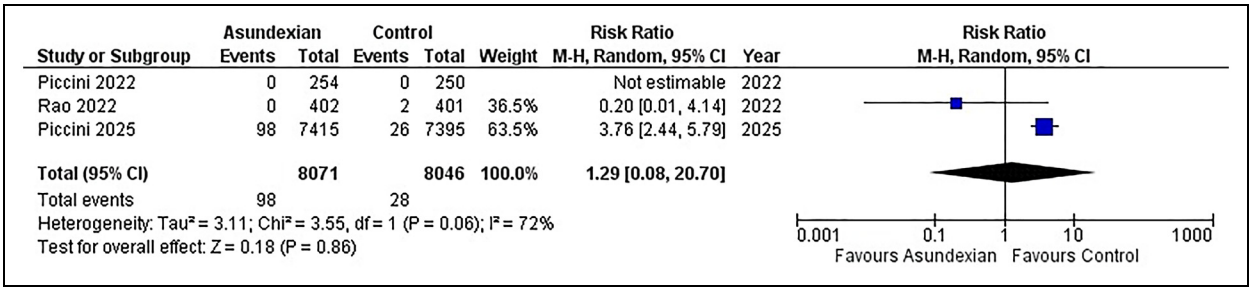

Figure 7. Forest Plot of Stroke or Systemic Embolism.

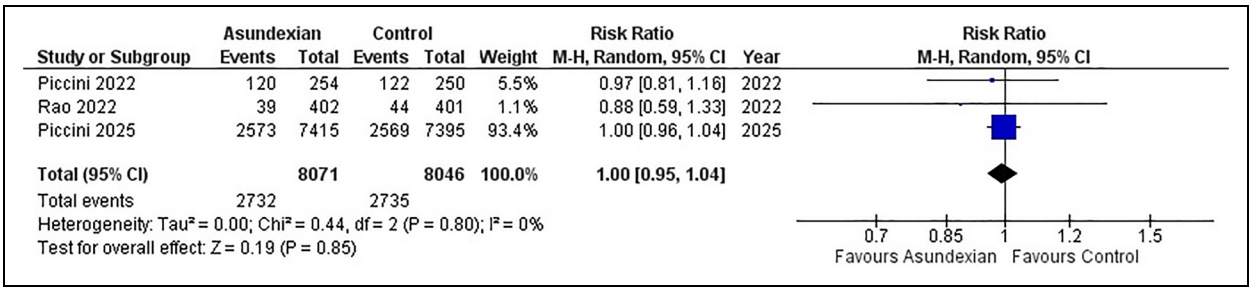

Figure 8. Forest Plot of Any Adverse Events.

Discussion

This systematic review and meta-analysis aimed to analyze the safety and efficacy of Asundexian, an oral novel factor XIa inhibitor in patients with cardiovascular diseases (CVDs) such as atrial fibrillation (Afib), myocardial infarction (MI), and ischemic stroke. By evaluating data from four different randomized controlled trials (RCTs), we focused on understanding if Asundexian can be considered a viable alternative to conventional antithrombotic agents, especially in high-risk populations where the balance between thrombosis and bleeding is very delicate. CVDs are the leading cause of morbidity and mortality worldwide, with the trend documented to have been worsening over the recent decade.[26] CVDs pose a significant risk of thromboembolic complications such as stroke, myocardial infarction, and systemic embolism across multiple populations.[27–29] Anticoagulation is pivotal in the prevention of these events, particularly in conditions such as atrial fibrillation where thrombus formation due to stasis of blood in the left atrium substantially increases the risk of stroke.[30] Standard agents used for anticoagulation against these risks include Aspirin, Vitamin K antagonists (VKAs) mainly warfarin, and DOACs such as apixaban, rivaroxaban, edoxaban, and dabigatran.[31] DOACs have emerged as a safer and more predictable alternative for VKAs with no need for strict monitoring, and a diminished risk for bleeding particularly intracranial bleeding.[31,32] Nevertheless, anticoagulation use, even with DOACs, is often constrained in clinical practice by apprehensions about heightened bleeding propensity, especially gastrointestinal hemorrhages,[32] renal dysfunction, and significant comorbidities that have conventionally challenged safe administration.[17]

Recently, a new oral anticoagulant has emerged namely Asundexian, which is a first-in-class Factor Xia inhibitor. Factor Xia serves as an appealing target for achieving anticoagulation since it selectively engages pathological thrombosis with limited involvement in physiologic hemostasis.[8,17] Inhibition of factor XIa allows dissociation of antithrombotic effectiveness and bleeding risk, which brings it closer to replacing traditional anticoagulants.[17] Moreover, individuals with congenital deficiency of factor XIa have demonstrated lower thromboembolic events with minimal bleeding complications,[17] further supporting the biological plausibility of this approach. Asundexian exhibits additional pharmacological advantages, including daily dosing convenience, prolonged half-life, negligible drug interactions, and predictable pharmacokinetics, thereby supporting adherence and clinical use.[8] Moreover, unlike warfarin, it does not require routine coagulation monitoring.[8] Preliminary Phase II data indicate that, compared to DOACs such as apixaban, bleeding rates may be similar or potentially lower, though its efficacy in completely preventing thromboembolic events, especially ischemic stroke in atrial fibrillation, remains uncertain.[33] Although this meta-analysis did not prioritize adverse effects, previous studies have identified gastrointestinal disturbances and headaches as notable concerns, meriting further investigation in subsequent pharmacovigilance research.[34]

The findings of our meta-analysis highlight both the therapeutic promise and inherent limitations associated with Asundexian. According to the evidence, the safety-efficacy profile is differentiated, with some favorable tendencies for bleeding risk reduction, but limited or even adverse findings in terms of thromboembolic event prevention, particularly in atrial fibrillation patients. Asundexian was associated with a significantly lower risk of ISTH-defined major or clinically relevant non-major bleeding in patients with atrial fibrillation, consistent with its mechanistic rationale.[17] This bleeding-sparing effect is debatably the drug's most alluring feature and may prove beneficial in populations with high bleeding risks or prior hemorrhage events.[35] However, this safety advantage was not uniformly observed across all cardiovascular subtypes. In patients with recent ischemic stroke or myocardial infarction, the bleeding risk reduction was less clinically substantial and statistically non-significant implying that the therapeutic window of Factor XIa inhibition may be pathology-dependent. This contrasts with the findings of a recent meta-analysis by Galli et al,[36] which suggested that factor XIa inhibition with Asundexian may reduce bleeding risks compared to enoxaparin or other DOACs in patients with ischemic stroke or myocardial infarction. This divergence may stem from differences in study inclusion criteria, dosing strategies, or comparator arms like our analysis included placebo and apixaban, while Gali's pooled comparisons involved enoxaparin or other DOACs or placebo in addition to antiplatelet therapy. This highlights the need for more granular subgroup analyses and head-to-head comparisons in future trials to clarify whether the safety benefits of FXI inhibition are consistent across diverse cardiovascular presentations.

From an efficacy perspective, Asundexian did not significantly reduce all-cause mortality or cardiovascular death in any of the subgroups analyzed. Moreover, in patients with atrial fibrillation, the risk of ischemic stroke and systemic embolism was significantly higher in the Asundexian group compared to the control, raising serious concerns about its independent utility in this setting. This concern is further supported by the early termination of the Phase 3 OCEANIC-AF trial, in which Asundexian was found to be inferior to apixaban in preventing stroke and systemic embolism in atrial fibrillation patients, a decision that demonstrates the potential limitations of FXIa inhibition in this specific population.[37] These results challenge the hypothesized advantages of Factor XIa inhibition and highlight the intricate pathophysiology of atrial fibrillation, where thrombogenesis is primarily driven by upstream coagulation cascades and atrial blood stasis.[38] The significantly increased risk of ischemic stroke with Asundexian in patients with atrial fibrillation (RR: 4.01) may be attributed to insufficient anticoagulation efficacy in this high-risk population. Unlike traditional anticoagulants that target thrombin or factor Xa, FXIa inhibition may not provide robust stroke prevention in patients with established atrial thromboembolic risk. Additionally, the PACIFIC-AF trial excluded patients with prior stroke or high CHA_2DS_2-VASc scores, potentially limiting the generalizability of its findings and resulting in underestimation of stroke risk in broader AF populations. The absence of a standard comparator like apixaban or warfarin in some studies may also skew interpretation. These findings suggest that Asundexian, while promising, may not be suitable as monotherapy for stroke prevention in AF until further evidence confirms its safety and efficacy in appropriately selected patients. The substantial heterogeneity was also observed in the analysis of ischemic stroke outcomes ($I^2 = 89\%$) and is likely attributable to clinical differences among study populations, particularly variations in baseline characteristics, comorbidities, and prior antithrombotic use. Additionally, differences in trial design, including inclusion criteria and outcome definitions, may have contributed. In stroke and myocardial infarction subgroups, no significant benefit nor a clear trend was favoring Asundexian for ischemic outcomes, although these results were less conclusive due to small sample sizes and wide confidence intervals.

Moreover, Asundexian showed no significant difference in the incidence of adverse events overall, supporting its potential as a well-tolerated therapeutic option. This is consistent with findings from early-phase studies, which demonstrated that Asundexian may provide enhanced safety relative to apixaban while maintaining a safety profile comparable to placebo.[33] Further supporting its pharmacological suitability, Kanefendt et al reported that Asundexian was well tolerated and exhibited complete bioavailability irrespective of tablet formulation, food intake, or gastric pH, making it a practical option for a broad range of patients with varying gastrointestinal conditions or dietary habits.[39] Collectively, these findings suggest that while Asundexian may reduce bleeding risk, its current formulation might lack sufficient thromboembolic protection in high-risk populations like atrial fibrillation. Thus, its use may be limited to specific cases where standard anticoagulants are ineligible or as part of combination therapies. Phase III trials with ischemic endpoints and direct comparisons to DOACs are warranted.

Although formal subgroup meta-analyses by condition (AF, stroke, MI) were not feasible due to the small number of studies per category, a brief narrative stratification can still offer clinical insight. For instance, patients with atrial fibrillation often have ongoing atrial thrombus risk requiring potent systemic anticoagulation, whereas those post-MI may have stent-related thrombosis dynamics, and post-stroke cases encompass varied pathologies (eg large-vessel vs cardioembolic infarcts). These clinical distinctions likely contribute to observed outcome variability. Future trials should pre-specify condition-based subgroups and ensure sufficient enrollment to support statistically

powered comparisons, enabling more tailored and reliable assessment of Asundexian's effects across different thrombotic conditions

Limitations

Several limitations of this meta-analysis should be recognized. First, the small sample size of only four RCTs reduces statistical power and may affect the external validity of results. Moreover, clinical heterogeneity is a potential concern, as the studies examined Asundexian across different cardiovascular conditions comprising atrial fibrillation, AMI, and ischemic stroke. We were unable to perform subgroup analyses due to the limited number of studies within each subgroup, which substantially reduces statistical power. Future research should prioritize adequately powered trials with pre-specified subgroups (eg, by population, comorbidities, baseline risk) and standardized reporting to enable representative and credible subgroup meta-analyses. Additionally, some outcomes such as ischemic stroke demonstrated significant heterogeneity, possibly arising from disparities in trial populations, follow-up periods, and comparator treatments (placebo vs active controls). Furthermore, the scarcity of long-term follow-up data constrains definitive assessment of Asundexian's prolonged efficacy and safety. Lastly, insufficient number of trials prevented a formal evaluation of publication bias.

Conclusion

Preliminary evidence from this meta-analysis suggests Asundexian's selective Factor XIa inhibition confers a favorable safety profile, with bleeding risks comparable to standard antithrombotic regimens across multiple cardiovascular indications. While no statistically significant reduction was noted in major cardiovascular outcomes. However, the observations remain provisional due to a limited number of trials, population heterogeneity, and insufficient follow-up duration. Larger well-designed randomized controlled trials with extended follow-up periods are needed to better assess the long-term effectiveness and safety of Asundexian in distinct cardiovascular patient populations.

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